

GUIDELINES AND FORMAT FOR CAPSTONE PROJECT

I. Introduction

Batangas Eastern Colleges would like its students to produce innovative researches that manifest their ability to undertake and present quality outputs and documentation that will contribute to the advancement of knowledge in their respective fields of study.

Capstone project is one of the requirements in different areas of specialization of the students of the Batangas Eastern Colleges in the College department in which students are expected to:

1. enroll, attend, defend and comply with the requirements;
2. organize and work in a team, share resources, tasks, and responsibilities;
3. follow rules and procedures, and plan activities to do the research;
4. consult regularly with the instructor/adviser; and
5. submit on time the research output.

Students shall comply with the standards set by BEC in system development and writing the documentation paper. They are expected to consult the office regarding the requirements for the development of the system.

The College Department approves the final paper and reserves the right to reject documents that depart from the guidelines specified in this manual. Students are advised to read this manual carefully and to pay close attention to the details of manuscript preparation, including the samples in the Appendices. In addition to structural formatting, they are also responsible for assuring that their manuscript follows conventional rules of grammar, punctuation, and spelling. They are required to seek the assistance of a professional statistician and grammarian to help them improve their work.

II. Trusts and Priorities

- a. *Utilizing information and technology-enabled business processes for competitive advantage.*
- b. *Bridging technical and management communities within an organization.*
- c. *Ensuring sound business operation through information system.*

III. Guidelines for Capstone Project Oral Defense

The guidelines provide details on the purpose, conduct, and conditions of the oral defense.

A. Purpose

1. To give an opportunity to developers to defend their study and reply to criticisms or challenges to the arguments present in their project while allowing members of the panel to clarify issues in the study;

2. To provide an opportunity for developers to interact directly to the panel by discussing the in-depth of their study;
3. To enable the member/s of the panel to provide the developers an agreed, single set of corrections or revisions required to be made before the award of qualification
4. To ensure excellence in the output of research study

B. Conduct

The developers are required to undertake an oral defense as part of their project study evaluation. Non-compliance will mean failure in the subject, Capstone Project. Any exemptions from undertaking the oral defense must be approved by the Dean of Colleges.

1. Stages of the Defense Process

First Stage: Title Defense

This includes chapters 1, 2, 3 and at least 30% of the developed system. The developers will defend the purpose and significance of the research on the chosen topic.

Second Stage: Final Output

The developers will present and defend their project.

The oral defense should be presented according to the following order:

1. A brief overview of the project
2. Relevant questions from the panel
3. Other questions / free discussions

Panel of examiners (group of teachers and practitioners) may give additional inputs/recommendations for further improvement of the study. Presenters should follow the scheduled date of defense and prepare a copy of research for the panel and adviser/instructor three (3) days before the date.

During the final defense, the developers and capstone project instructor/research adviser/facilitator have to write down the suggestions and recommendations of the panelists on the table matrix given before the Final Defense.

2. Schedule

- a. The oral defense should be scheduled once the group of students are nominated for oral defense. Nomination should come from the research instructor. The instructor shall also nominate the members of oral examination panel with the recommending approval of the Department Head.
- b. The manuscript should be submitted to their capstone project instructor for checking and evaluation. Oral defense will be set for one (1) to two (2) weeks after the manuscript has submitted.

- c. If the members of the panel found out that the project is considerably substandard, the oral defense will be rescheduled.
- d. The oral defense will be scheduled depending on the availability of the members of the panel and the need to finish the presentations.

3. Participants

The oral defense shall be attended by:

- a. The Proponent/s
- b. The Capstone Project Instructor/Adviser
- c. Members of the Panel
- d. The Chair
- e. Guests/Audience at the discretion of the Chair
- f. Dean of Colleges

The chairperson/lead panelist shall be drawn from the BEC faculty who coordinates, oversees and facilitates the oral defense. The instructor/adviser of the researchers who will defend their study shall not be appointed as panel and neither shall anyone who has a personal relationship with them.

The chair is responsible for ensuring that:

- a. process is made clear to all participants and progresses smoothly
- b. the members of the panel ask questions about the research study and discuss other matters related to the research
- c. questions to be asked shall technically relevant to the research study
- d. there is adequate opportunity for the researchers to defend their work and respond to any questions or criticism
- e. researchers are fairly treated and not subjected to unjustifiable stress
- f. agreement about the defense result is clearly communicated

Member/s of the panel shall be approved by the Dean of Colleges and the Vice President for Academic Affairs upon the recommendations of the Research Instructors.

The panel's decision and any feedback from the members of the panel must be indicated in the table matrix given to the researchers and research instructor. The chair and the member/s of the panel who attended the Pre-Oral Defense will be the same panel who will examine the output of the students' research study in the Oral/Final Defense.

4. Venue

On the day of the defense, the following should be observed:

A. OFFLINE

- The developers are expected to be at the venue 30 minutes before the time.

- The developers should ensure that multimedia devices are available during their defense schedule.

B. ONLINE

- The developers shall send the link to the concerned persons the day before the defense.
- The developers are expected to log in 30 minutes before the time.
- The developers shall ensure the availability of all the materials needed.

5. Duration

The intention of the oral defense is to be a stimulating and constructive intellectual exchange. A typical oral defense session will take up at least one (1) hour and 30 minutes, which includes:

1. Proponent's research presentation	30 minutes
2. Question and answer	30 minutes
3. Post-conference after the oral defense	30 minutes

C. Conditions (Face-to-Face)

- The project which requires statistics needs to be consulted to the statistician;
- All manuscript should be checked by a grammarian;
- Statistician and grammarian are approved by the Dean of Colleges upon the recommendations of the research instructors/advisers;
- Developers who will conduct research on other campuses will bring a referral letter signed by the school librarian
- Developers should wear proper school uniform and school ID whenever they do their research outside the school
- The attire for the oral defense is corporate.

IV. Conducting Oral Defense

Every oral defense shall have the parts as given below.

I. Opening Prayer.....Candidate

II. Greetings and Introductions

GreetingsPanel on Oral Examination (POE) Chair

Introduction of the CandidateResearch Adviser

Introduction of the Panel on Oral Examination...POE Chair

III. Announcement of the Rating SystemPOE Chair

IV. Oral Examination Proper

Brief Presentation of the StudyCandidate

Question-and-Answer Phase

(General to specific questions)Panel on Oral Examination,
Candidate and Adviser
(Recommendations/Suggestions to improve the study)
.....Panel on Oral Examination

V. Concluding RemarksPOE Chair

VI. Deliberation (Candidates and guests leave the room)..POE Chair

VII. Announcement of Results.....POE Chair

V. Grading System

Criteria for Presentation and Manuscript	Weight
I. Final Defense	
Presentation: Style of Presentation and Oral articulation	20%
Ability to answer questions and Mastery of the topic	40%
II. Manuscript	
Contribution to the field of specialization/discipline	10%
Substance and Content (Logical organization, Accuracy of the research process, Accuracy and relevance of works cited depth of discussion)	20%
Form and Writing Style (use of proper format; proper application of rules of style and grammar, scholarly language).	10%
Total	100%

OUTPUT

Criteria	Weight
1. Functionality – performance of the system for the intended tasks and meets specified requirements.	30%
2. Usability – It pertains to user experience, including the interface design, ease of use, and accessibility.	25%
3. Completeness - fully developed and all key features and functionalities are implemented.	20%
4. Innovation - originality and creativity in solving problems or implementing unique features.	15%
5. Teamwork - collaboration among team members and contributions toward the system's development.	10%
Total	100%

VI. Statistical and Editorial Consulting

The developer/s shall be expected to ensure appropriateness of the statistical treatment used in the manuscript. He may consult research and statistical experts within and outside the Colleges as recommended by the Capstone Project Instructors and approved by the Dean of Colleges.

The editing of the manuscript is a requirement to check for grammatical and mechanical errors. A certification from a professional statistician and editor that the manuscript has been statistically treated and thoroughly edited shall be included in the Appendices.

VII. Required Number of Final Copies

The developer/s shall be required to submit two (2) copies of the manuscript, which will be distributed to the Library and Office of the Dean. An additional copy is optional when the proponents wish to have their own.

VIII. Payment

An oral defense fee per student/researcher will be paid directly to the cashier to cover the following expenses;

- a. Honorarium for Title and Final Defense
- b. Statistician and Grammarian Fee
- c. Internet Subsidy (Online)/Food (Face-to-Face)

BSIS Capstone Format

I. I Parts of a Research Paper

The parts of the research paper for a pre-oral and oral defense shall be as follows:

Preliminaries

- Title Page
- Certification and Approval Sheet
- Abstract
- Acknowledgment
- Dedication
- Table of Contents
- List of Tables
- List of Figures

Note: Pagination of preliminaries shall be in lowercase Roman numerals placed at the bottom center of the page.

Chapter 1 INTRODUCTION

- Project Context (Background of the Project)
- Objectives of the Project
- Significance of the Project (This must be anchored to the institution's research thrusts and priorities)
- Scope, Delimitation, Limitation of the Study
- Definition of Terms

Chapter 2 REVIEW OF RELATED LITERATURE

- Literature Review
- System Review
- Theoretical or Conceptual Framework
- Project Paradigm

Chapter 3 METHODOLOGY

- Materials and Tools (Software and Hardware)
- Research Design
- Development Methodology
- Population of the Study
- Sampling Design
- Data Collection Instrument
- Validation of the Questionnaire
- Statistical Treatment

- Ethical Consideration

Chapter 4 PRESENTATION, ANALYSIS AND INTERPRETATION OF DATA

- Presenting, analyzing, interpreting and supporting the data

Chapter 5 SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

Bibliography

Appendices

II. Manuscript Preparation

In the preparation of the manuscript, care shall be taken to ensure the following specifications:

Paper and Printing Specifications

Print the final copy using only one side of a standard-sized BEC template (**8.5 x 11 inches**)

Font Specifications

Use **12-point Times New Roman** font throughout the paper except when compressing a large table, use 10-point, title is 14-point.

Spacing

Double space the text; single-space long quotations, table and figure captions, and similar special materials (e.g., table legend). Type triple-spaced from the top margin, centered, boldfaced, and in ALL CAPS all headings of the preliminary section (except the Title Page) and all chapter titles.

Margins

Leave 1.5 inches for the left margin to allow for binding and trimming and 1 inch for the top, bottom, and right margins.

Page Numbering

Paginate the preliminaries by using lowercase Roman numerals at the center, bottom margin.

Use Arabic numerals to paginate the text, references, and appendices. Number all the pages consecutively starting at number 2 on the second page of the first chapter. The first page of each chapter, though counted, shall not be numbered.

Type the page number at the bottom center of the paper (i.e., tables and figures).

Preliminaries

The preliminaries include the title page, certification and approval sheet, abstract, acknowledgment, dedication, table of contents, list of tables, and list of figures.

Cover and Title Pages

Follow the instructions on the attached sample-form pages. Note which items are in ALL CAPS and which are not. Note the approximate size of the print (***see Appendix 1&2: Cover Page and Title Page.***)

Certification and Approval Sheet

The CERTIFICATION contains the approval and acceptance of the research adviser as the partial fulfillment of the requirements for the degree.

The **APPROVAL** contains the formal approval or commendation of the Panel on oral examination headed by a chairperson). (***See Appendix 3: Approval Format.***) Refer to the attached sample for the wording and format that shall be followed faithfully. Include this sheet in each final copy of the research.

Abstract

This is a brief and concise descriptive summary of study containing the introduction, methodology, results and discussions. It must contain the researchers' name as it appears on the title page, the degree, the date (last month of the semester in which the student completes the degree), title of the research (wording exactly to agree with the Title page), and name of the research adviser (First Name, Middle Initial, Surname, followed by honorifics. – Not all in uppercase).

The abstract should not be less than **200 words and not more than 250 words**, and should be typed double-spaced and preferably with two to three keywords. Normally the abstract does not include any reference to the literature. See example in appendix 4.

Acknowledgment

Type triple-spaced from the top margin, centered, boldfaced and in ALL CAPS the heading **ACKNOWLEDGMENT**. Double-space the text that begins three single spaces from the heading. Third personal pronoun should be used and names of researchers should be arranged in First name M.I. Lastname.

Table of Contents

The Table of Contents comes after the acknowledgement. Type triple-spaced from the top margin, centered, boldfaced, and in ALL CAPS the heading **TABLE OF CONTENTS**. Type double-spaced from the heading. It shall list all elements of the preliminaries, chapter titles,

main headings, references, and appendices. The beginning page number of each section is indicated along the right margin. The numbering of the chapters and the wording, capitalization, and punctuation of titles and headings, shall be exactly the same as they are in the text.

Example:

	Page
TITLE PAGE	i
CERTIFICATION AND APPROVAL SHEET	ii
ABSTRACT	iii
ACKNOWLEDGMENT	iv
DEDICATION	v
TABLE OF CONTENTS	vi
LIST OF TABLES	vii
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CHAPTER	
1 INTRODUCTION	
Project Context	
Objectives of the Project	
Scope, Delimitations and Limitations of the Project.	
Significance of the Study	
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2 REVIEW OF RELATED LITERATURE	
Literature Review	
System Review	
Theoretical or Conceptual Framework	
Research Paradigm	
3 METHODOLOGY	
Research Design	
Materials and Tools (Software and Hardware)	
Development Methodology	
Population of the Study	
Sampling Design	
Data Collection Instrument	
Validation of the Questionnaire	
Statistical Treatment	
Ethical Consideration	

4 PRESENTATION, ANALYSIS AND INTERPRETATION OF DATA

Presenting, analyzing, interpreting and supporting the data (PAIS)

Objective 1

Objective 2

5 SUMMARY OF FINDINGS, CONCLUSIONS AND RECOMMENDATIONS

Summary of Findings

Conclusions

Recommendations

REFERENCES

(APA latest edition)

APPENDICES

A – Communication Letters

B – Survey Questionnaire

C – Interview Transcripts

D – Functional Testing Results

E – User Acceptance Testing Results

F – ISO 25010 Evaluation Results

G – Screenshots

H - Documentation

I – Source Code Repository Link

J – Grammarian Certificate

K – Statistician Certificate

L – Curriculum Vitae

1. INTRODUCTION

This chapter serves as a backgrounder for readers to have an overview of the study even without prior reference to other publications on the topic. It is in this section where it discusses the overview of the research problem, define major concepts, discuss the factors that lead the proponents to the conceptualization of the problem.

It should contain 2 to 3 pages only, and wordings should be 90 / 10 or 80 / 20. 90 or 80 percent should be own words from researcher/s and 10 or 20 percent should be cited from related studies and/or literature.

The content of the introduction should discuss the general topic, broader context, and relevance of the study, gradually leading the reader to the specific project. Think of it as moving

from Global Context → Industry Context → Information Systems Context → Research Need without yet discussing the specific organization.

This can be organized as **paragraph 1** - General Background of the Topic that introduce the broader field related to the project. **Paragraph 2** - Importance of Technology in the Domain which discuss how technology is being used in the specific area. **Paragraph 3** - Existing Challenges, discuss common problems experienced by organizations related to the topic. **Paragraph 4** - Relevance of Information Systems, discuss how an information system can address these issues. **Paragraph 5** - Transition to the Project Context, lead the discussion toward the target organization.

Project Context

- a. The Project Context presents the environment and circumstances that justify the development of the proposed system. It provides a clear understanding of the existing situation, identifies operational problems and gaps, and explains the need for the proposed solution.
- b. The discussion shall begin with the **Organizational Profile**, which includes a brief history of the organization, its vision and mission, services offered, and organizational structure. This should be followed by a description of the Existing Business Process, explaining the current workflow and whether the activities are performed manually, through an existing system, or a combination of both.
- c. Researchers shall then discuss the **Problems and Gaps identified in the current process**, such as inefficiencies, delays, data inaccuracies, security concerns, lack of integration, or limited automation. These issues should be supported by data gathered through interviews, observations, documents, or other relevant sources.
- d. The section shall conclude with the **Need for the Proposed System**, explaining how the proposed solution will address the identified problems, improve existing processes, and provide benefits to the organization and its stakeholders.

Objectives of the Project

General Objective, this is a statement that states what the research or thesis project is trying to accomplish. This also contains specific objectives which that are

- a. statements that try to achieve the general objective. Specify the things that would be done to accomplish the general objective.
- b. Objectives are statements of WHAT the project is expected to accomplish. Each objective should be stated to describe what is to be done. Since objectives are associated with action, they usually start with action verbs.

Stating project objectives:

The study aims to...

develop	conduct	assess
design	implement	enhances
produce	train	strengthen
acquire	improve	evaluate

- c. A satisfactory objective should be Specific, Measurable, Attainable/Achievable, Realistic, and Time-bounded. It should be stated to include the following information:
- an action verb;
 - the outcome to be accomplished;
 - the time-frame the outcome is to be accomplished; and
 - the criteria or conditions for measuring the accomplishment.
- d. Few objectives should be considered in a proposal and they should be arranged in their order of importance. It is more likely that few objectives can be successfully accomplished given the available resources.
- e. Objectives must always relate to the expected outcomes or project outputs. Moreover, objectives determine the methodology – how each objective is to be accomplished.
- f. Objectives are normally classified as general and specific. A general objective states what the research or thesis project is trying to accomplish. Specific objectives are statements that try to achieve the general objective.
- g. Analysis of the objectives:

Criteria	Justifications
Specific?	YES! Because the objectives state particular, precise and definite details about the project
Measurable?	YES! Because the objectives state weighable (80%) indicator to measure the success of the project
Attainable?	YES! Because the procedures, function and integration of GRS and POS is feasible and within reach
Realistic?	YES! Because the objective states practical and viable methodology
Time-bounded?	YES! Because the objective states the time when the system will be accomplished

Example of drafting the objectives

General Objective (Overall goal of the project)

To develop a Web-Based Maintenance Management System for Batangas Eastern Colleges.

Specific Objectives (System Development Objectives)

1. To develop a defect reporting module that includes:
 - 1.1. Report submission
 - 1.2. Photo attachment
 - 1.3. Report tracking
2. To develop a maintenance management module that includes:
 - 2.1. Work order generation
 - 2.2. Technician assignment
 - 2.3. Maintenance scheduling
3. To evaluate the developed system using ISO 25010 in terms of:
 - 3.1. Functional Suitability
 - 3.2. Usability
 - 3.3. Reliability
 - 3.4. Security
 - 3.5. Performance Efficiency

Scope, Delimitation and Limitations

- a. It states the limitations of the study. It specifies the topic and its participants or subjects of the study. This section of your study launches the limits of the process in which your study will be conducted. Discuss the variables that you will examine and the method that you are going to use. This will also mention the number of participants, the place where will get the data, the instrument to be used, and the time frame for completing your study.
- b. Determining the scope of the study will declare the feasibility of doing the investigation. It will set the applicability of findings. Only those stated in the scope shall cover the credibility and generalizability of results.
- c. It should contain three (3) paragraphs that discuss each separately, one paragraph for the Scope, another one paragraph for the Delimitation and one for limitations
- d. For the **Scope** this should discuss the features included in the system, for **Delimitations**, it discusses the features intentionally excluded and for the **Limitations**, it discusses the constraints beyond researchers' control for example internet dependency, availability of respondents, testing environment limitations.

Significance of the Study

- a. It answers the questions “Why is your research study important/ significant to your study. Describe general contribution of the project to new knowledge, society and or to development in general. It points out the advantages that will come from the results. It emphasizes the concerned population to which the

findings will be favorable. Usually, the advantages that may come from your study should be detailed in this section.

- b. This should discuss where study will be significant. It should be aligned to the institutional trusts and priorities as well as to the sustainability development goal.

Definition of Terms

- a. Only the variables or key terms included in the study shall be defined.
- b. There is no need to define all the terms operationally. Only terms with no available published definitions, terms that were used contextually in the study, and terms that entail the output of the study must be defined operationally.
- c. The terms shall be sequenced in **alphabetical order**.
- d. If the terms consist of more than one word, **capitalize every major word**.
- e. The term to be defined shall be indented, typed in a bold face, and punctuated with a period. Two spaces shall separate the period from the definition
- f. Each definition shall be expressed in a complete sentence, NOT a phrase.
- g. Symbols and abbreviations shall be enclosed in parentheses following their spelled-out nomenclatures.

2 REVIEWS OF RELATED LITERATURE

This chapter presents the review of literature and studies that have bearing on the present study, its synthesis, and the conceptual framework of the study. These should be organized by themes.

Literature Review

Topic/ Theme 1. Digital Transformation

Topic/ Theme 2. Maintenance Management Systems

Topic/ Theme 3. Work Order Management

Topic/ Theme 4. Mobile Reporting Systems

2.1. For every theme there should be at least fifteen authors of foreign and local studies. The Review of Literature and Studies shall be organized thematically; therefore, the subheadings will be based on the themes/topics. The last paragraph in each theme/ topic may express the synthesis. The last two paragraphs of the literature review section may communicate the general synthesis.

2.2 Except for classical theories and any other relevant literature, all the literature and studies included in the review shall have been published or written (if unpublished, such as theses

and dissertations) at least five (5) years before the conduct of the current study, and ten (10) years for conceptual theory and related literature.

3.3 Except for exploratory studies, the **minimum number of literature and studies to be reviewed** shall be pegged at fifteen (15) or more.

System Review

The Review of Related Systems **presents existing systems, applications, or technologies that are similar to the proposed project.** This section helps researchers understand how other systems address similar needs and provides ideas that may be adopted or improved in the development of the proposed system.

For each related system, the researchers should briefly describe its **purpose and how it is used.** The **major features and functionalities** of the system should be discussed to show what it can offer to its users. The researchers should also **identify the strengths of the system**, such as its efficiency, ease of use, reliability, or unique capabilities, as well as **its weaknesses** or limitations. Through this analysis, the researchers can determine opportunities for improvement and justify how the proposed system can provide a better or more suitable solution to the identified problem.

Theoretical or Conceptual Framework

Present specific and well-defined constructs, assumptions, expectations and beliefs that support the research study. **The Theoretical Framework** identify the theories supporting the study such as Technology Acceptance Model (TAM), Diffusion of Innovation Theory, DeLone and McLean IS Success Model and others. The **Conceptual Framework** illustrates relationship among variables.

Research Paradigm

A diagram that illustrates the relationship of the variables of the study. This may take the form of (1) input-process-output; (2) system framework; (3) SDLC framework

METHODOLOGY (Here the discussion in System Analysis will be applied)

Research Design

This section states the research design which refers to the overall strategy that you choose to integrate the different components of the study in a coherent and logical way, thereby, ensuring you will effectively address the research problem; it constitutes the blueprint for the collection, measurement, and analysis of data.

Explain the research design you will use. Discuss your reasoning for choosing either qualitative or quantitative research or for using a blended approach or descriptive, correlational, experimental, developmental, or any alternative method your specific field recognizes.

Materials and Tools

This section states what hardware and software tools you will use in the completion of the system/model/prototype.

Hardware Requirements *(includes in the development, deployment and testing)*

In this section the hardware to be used will be discuss.

Example:

Desktop. It is used to perform the development of mobile application software. It must have a good specification which can run the used software such as Visual Studio Code and TensorFlow. Only one computer is used during the development of the said web based.

Table 1. Desktop Minimum Specification

Processor	Processor Intel ® core ™ i3-4030 CPU @ 3.30GHz
Memory	4.00 Gigabytes
Hard Disk Drive	500 Gigabytes
System Type	32-bit Operating System

Android Phone. It is the most important hardware device for the implementation of the project. Developing a responsive web-based system will help everyone to access the recommender system through their mobile phone without using a high-end device.

Table 2. Android Phone Minimum Specification

Processor	1.95 GHz Snapdragon 439 Octa core
Memory	2 GB Ram
Internal Storage	2 Gigabytes
Browser	Chrome

Software

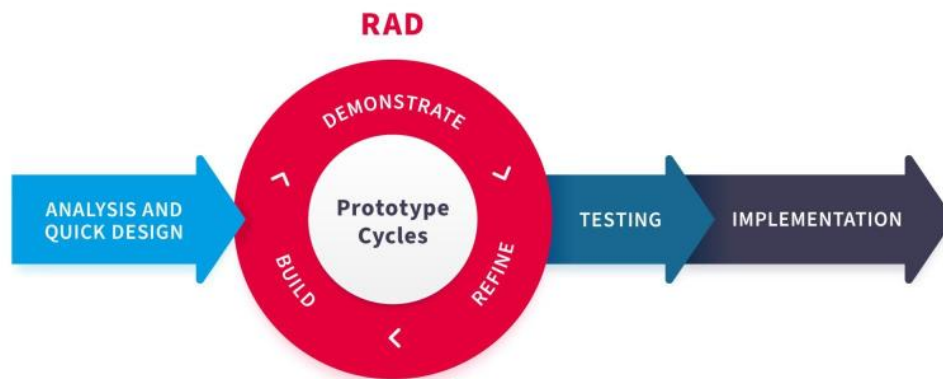
This will discuss the different software to be use in the project such as Database, IDE GIS Tool if applicable, Browser and others. Identify the software and explain why and how it will be used in the project.

Development Methodology (if applicable)

This section discusses the researchers' developmental methodology as the algorithm or data model integration in the software design. The models specify the various stages of the process and the order in which they are carried out.

Describe what Software Development Life Cycle (SDLC) you will follow. E.g. waterfall methodology, iterative model, continuous model, agile methodology, lean development, rapid

application development, prototype model, feature driven, scrum, or any alternative method you know.



Source: Website Name/Article Name (if any)

Figure 1. Rapid Application Development

You may add figures in your discussion. Label of tables and figures are centered, bold and size 10. For tables, the label must be placed above the table while for figures it must be placed below the figure. All figures must be cited within the text. The word Figure must be used instead of figure or Fig when referring to the Figure. If figures are small, they can be arranged differently.

All tables must be cited within the text. For example, Table 1 should be referred to as Table 1 and not table 1 or anything else. Font size for text within tables should normally be at 10 pt. If majority of the tables are small in width, they can be left-aligned but this must be consistent throughout the paper.

Explain what actions you will take **on each phase of the model**. If your study requires you to add **system/project design**, you may add discussion on models such as System Architecture, Use Case Diagram, Context Diagram, Data Flow Diagram, or any other diagrams/models related and will support the discussion of your methodology.

Example Content of the Development Methodology

Overview of the Methodology

Methodology Diagram

Phase 1

Phase 2

Phase 3

Phase N

PROJECT DESIGN

Presents all the diagram needed in the project

Population of the Study

The participants are described in terms of the total population and representative sample chosen. The researchers shall include a discussion about the tool used in determining the sample size, the sampling technique, and the procedure used in selecting the respondents.

Tabular presentation of the respondents would only be applicable to samples with grouping/ strata.

Data Instrumentation

The research instrument which either can be a questionnaire, test interview, observation schedule, or rating scale must be described in how it is designed and used by the researcher. CAS (Construction, Administration, and Scoring) will be presented in this part. If there will be two (2) or more instruments used in the study, each instrument will be presented per section with CVAS.

Validation of the Questionnaire (if only applicable)

The questionnaire designed for the study was subjected to a validation process for face and content validity.

Testing and Evaluation Procedures

This section describes the procedures used to verify the functionality, performance, and quality of the developed system. It explains how the researchers ensured that the system met its specified requirements and achieved its intended objectives.

The Testing Procedures shall discuss the methods and activities conducted to determine whether the system functions correctly and performs as expected. Researchers shall identify and describe the testing techniques employed, such as Unit Testing, Integration Testing, System Testing, User Acceptance Testing (UAT), or other applicable testing methods. For each testing activity, the objectives, test scenarios, test cases, expected results, actual results, and corrective actions taken should be presented and discussed.

The Evaluation Procedures shall explain how the developed system was assessed by its intended users, experts, or other qualified evaluators. Researchers shall identify the evaluation model, framework, or criteria used in assessing the system, such as ISO 25010, Technology Acceptance Model (TAM), System Usability Scale (SUS), or other relevant standards. The discussion should include the respondents involved, the evaluation instrument used, the administration of the evaluation process, and the criteria for interpreting the results.

Researchers shall also discuss the procedures for collecting, analyzing, and interpreting the testing and evaluation data. The results of the testing and evaluation should provide evidence that the developed system is functional, reliable, usable, and capable of meeting the needs of its intended users.

For system development projects, it is recommended that the Testing and Evaluation Procedures include:

Testing Procedures

- ✓ Unit Testing
- ✓ Integration Testing
- ✓ System Testing
- ✓ User Acceptance Testing (UAT)

Evaluation Procedures

- ✓ Respondent Groups (e.g., End Users, IT Experts, Administrators)
- ✓ Evaluation Instrument
- ✓ Evaluation Criteria

ISO 25010 Quality Characteristics:

- ✓ Functional Suitability
- ✓ Performance Efficiency
- ✓ Compatibility
- ✓ Usability
- ✓ Reliability
- ✓ Security
- ✓ Maintainability
- ✓ Portability

Statistical Treatment

The statistical tools used to answer the specific problem posited in Chapter 1 must clearly describe and explain the process and treatment of data, relevance and usefulness in the study.

Identify and justify the statistical treatment per objective. Present and justify the scale of values used and the descriptive equivalent ratings, if any. In case of the IT project, e.g. software/systems development, present and discuss the software/systems development process used. Include justification why such is used.

Ethical Considerations

- ✓ This includes the discussion about respondents' voluntary participation, informed consent, anonymity, confidentiality, the potential for harm, and results in communication.

4 RESULTS AND DISCUSSIONS

Presentation, Analysis, and Interpretation of Data

Chapter 4 presents the findings of every problem stated in Chapter 1. The findings must be presented directly. There is no need to mention the numerical data as they are already shown in the table. Items/ indicators must be presented, analyzed, interpreted, and supported by appropriate literature. Output of the study should be included in chapter 4.

Example:

As can be seen on the table, students are highly aware that higher education develops their communication skills by exposing them to realistic learning experiences. It reveals that the respondents believed that when they pursue higher education after SHS graduation, they would be given various opportunities that will enable them to develop effective communication skills needed to be productive citizens of a highly globalized society.

It indicates that higher education provides learning experiences for students that are contextualized to the needs of society. It conforms to the idea of Hoenick (2019) that pursuing a college education will provide students the opportunities to be trained as individuals who can express themselves clearly and communicate more effectively with others.

The number of items to be interpreted would be dependent on the number of indicators presented in the table. For a table with ten (10) items, at least the top three (3) and bottom three (3). All these items must be presented, analyzed, interpreted, and supported by appropriate literature separately.

Always interpret the composite mean at the last part of each subheading/ section as they suggest the overall impression of the data trend.

There is no need to show the ranking on the tables. Instead, arrange the items according to the weighted mean.

Example:

Table 6
Extent of Students' Awareness to Higher Education

Items	WM	VI
1. Develops students' communication skills by exposing them to realistic learning experiences	3.54	HA
2. Ensures that specialized and applied skills will be developed in offered Degree programs relevant to HUMSS strand	3.52	HA
3. Provides opportunities for personal development	3.43	MA
4. Guarantees the development of students' professional and academic competence	3.41	MA
5. Ensures the adequacy of the mastered basic competencies such as literacy, numeracy, and problem-solving	3.26	MA
Composite Mean	3.43	MA

Legends: **WM** – Weighted Mean **VI** – Verbal Interpretation **HA** – Highly Aware **MA** – Moderately Aware

Tables and Figures

- 4.1. Type every table its number and title.
 - 4.2. Type the table number at the center and triple-spaced from the last line of the immediately preceding paragraph.
 - 4.3. Type the title of the table at the center, title case, and single-space from the table number, and double-spaced to the table.
 - 4.4. Type triple-spaced from the last line of the immediately preceding paragraph its figure number and title.
 - 4.5. Type the title of the figure at the center, title case, and single-space immediately after the figure number.
 - 4.6. The numbering of tables and figures shall be continuous.
 - 4.7. If the table is still too large to fit a single page, reduce the Times New Roman font to a minimum of 8 points.
 - 4.8 If the table does not fit the page even after the font-size reduction, continue the rest of its part to the next page, but the cut part shall bear the subtitle, 'Continuation' flush with the left margin (e.g., Continuation of Table 14). The cut part/s shall also show the column headings and the legend (if any) like those in the first part of the table.
 - 4.9 Put two or more small tables or figures on a single page if the page is big enough to accommodate them all.
 - 4.10 Have small tables and figures appear on the same page along with the text; however, leave three spaces between every table/figure and the texts above and below it.
 - 4.11 The width of the table shall be dependent on the length of its title.
- ✓ For the presentation of the IT project and its discussions, the following may be used:
- In the case of an IS Plan, the IS Plan may follow any of the established frameworks, such as that of the National Computer Center.
 - For software systems development, discussion shall include but not limited to:
 - Description of the Project
 - Requirements (Functional and Non-functional)
 - Design of Software, Systems, Product, and/or Processes encapsulated using any appropriate CASE tools
 - Development and Testing, where applicable
 - Implementation Plan (Infrastructure/Deployment) where needed
 - Implementation Results, where applicable

5 SUMMARY OF FINDINGS, CONCLUSIONS AND RECOMMENDATIONS

Chapter 5 shall only have the following three subheadings: Summary of Findings,

Conclusions, and Recommendations.

The **Summary of Findings** must be consistent with Chapter IV. Thus, it must be **presented per section** anchored to the subtitles found in the preceding chapter. No data shall be observed in the presentation of the summary.

Conclusions

These are generalized statements from a micro to a macro level based on the answers to the general problem and each of the specific problems/objectives. General inferences are presented which are applicable to a wider and similar population.

The Conclusions must be anchored to the Research Objectives. Thus, **the number of conclusions in the study shall depend on the number of Objectives of the Project**. This should contain 1 sentence only.

Recommendations

These should be based on the findings and conclusions. Recommendations should be feasible, workable, flexible and adaptable in a non-technical language and may include suggestions for further studies.

In-Text Citation

Use the American Psychological Association (APA) style in the in-text citation of research of all programs. The APA style follows the author-date method of in-text citation. This means that the author's last name and the year of publication for the source shall appear in the text, e.g., (Jones, 2006) or 'According to Jones (2006)...'. A page number of the source, however, shall be included in the in-text citation in case of direct quotations, e.g., (Jones, 2006, p. 1-99).

The author-date method of the citation requires that the surname of the author (do not include suffixes such as Jr.), and the year of publication be inserted in the text at the appropriate point.

Examples:

One Work by One Author

- Kessler (2003) found that among epidemiological samples
- Early onset results in a more persistent and severe course (Kessler, 2003)
- " In 2003, Kessler's study of epidemiological samples showed that ...
- Among epidemiological samples, Kessler (2003) found that The study also showed that there was a high rate of comorbidity(Kessler 2003).

One Work by Multiple Authors

When a work has two authors cite both names every time the reference occurs in text. When a work has three, four, or five authors, cite all authors the first time the reference occurs. In subsequent citations, include only the surname of the first author followed by et al., {not italicized and with a period after al) and the year if it is the first citation of the reference within a paragraph.

- Kisangau et al. (2007) found [Use as subsequent first citation per paragraph thereafter.
- Kisangau et al. found [Omit yeai from subsequent citations after first nonparenthetical citation within a paragraph.

Exception: If two references of more than three surnames with the same year shorten to the same form), cite the surnames of the first authors and of as many of the subsequent authors as necessary to distinguish the two references, followed by a comma and et al.

Examples:

- Ireys, Chernoff, DeVet, and Kim (2001) and Ireys, Chernoff, Stein, et al. (2001)
- ...as Kurtines and Szapocznik (2003) demonstrated
- ...as has been shown (Joneskog&Sorbom, 2007)
- Kosslyn, Koenig, Barrett, Cave, Tang, and Gabrieli (1996)

If a reference list includes publications by two or more primary authors with the same surname, include the first author's initials in all text citations, even if the year of publication differs.

- Light, I. (2006). Deflecting innmigration: Networks, markets, and regulation in Los Angeles. New York, NY: Russell Sage Foundation.
- Light, M. A., & Light, I.H. (2008). The geographic expansion of Mexican immigration in the United States and its implications for local law enforcement. Law Enforcement Executive Forum Journal, 8,73-82.

When a work has no identified author; cite in the text the first few words of the reference list entry (usually the title) and the year. Use double quotation marks around the title of an article, a chapter, or a web page, and italicize the title of a periodical, a book, a brochure, or a report.

References

The American Psychological Association (APA) latest edition shall be used. The References should contain the author's name, date of publication, the title of the work, and publication data. The used references are classified as Books, Unpublished Materials, Journals/Periodicals, and

Electronic Sources. Using the justify alignment, the first line of information is aligned left and the next lines are 0.5 indented.

The author's name flush left, title case, single-spaced, and double-spaced to the next reference.

Example :

REFERENCES

A. Books

Format:

Author's last name, first initial. (Publication date). *Book title*. Additional information. City of publication: Publishing company.

Examples:

Allen, T. (1974). *Vanishing wildlife of North America*. Washington, D.C.: National Geographic Society.

Boorstin, D. (1992). *The creators: A history of the heroes of the imagination*. New York: Random House.

Nicol, A. M., & Pexman, P. M. (1999). *Presenting your findings: A practical guide for creating tables*. Washington, DC: American Psychological Association.

Searles, B., & Last, M. (1979). *A reader's guide to science fiction*. New York: Facts on File, Inc.

Toomer, J. (1988). *Cane*. Ed. Darwin T. Turner. New York: Norton.

Encyclopedia & Dictionary

Format:

Author's last name, first initial. (Date). Title of Article. *Title of Encyclopedia* (Volume, pages). City of publication: Publishing company.

Examples:

Bergmann, P. G. (1993). Relativity. In *The new encyclopedia britannica* (Vol. 26, pp. 501-508). Chicago: Encyclopedia Britannica.

Merriam-Webster's collegiate dictionary (10th ed.). (1993). Springfield, MA: Merriam-Webster.

Pettingill, O. S., Jr. (1980). Falcon and Falconry. *World book encyclopedia*. (pp. 150-155). Chicago: World Book.

Tobias, R. (1991). Thurber, James. *Encyclopedia americana*. (p. 600). New York: Scholastic Library Publishing.

B. Unpublished Materials

Format

Author, A. A. (Year of creation). *Title of work* (Type of thesis). Name of Institution, Location.

Format - Thesis from a database

Author, A. A. (Year of creation). *Title of work* (Type of thesis). Available from Database. (Accession or order no.)

Format - Thesis from the Web

Author, A. A. (Year of creation). *Title of work* (Type of thesis, Name of Institution). Retrieved from <http://xxxxx>

C. Journals/Periodicals

Magazine & Newspaper Articles

Format:

Author's last name, first initial. (Publication date). Article title. *Periodical title*,
volume number(*issue number if available*), inclusive pages.

Note: Do not enclose the title in quotation marks. Put a period after the title. If a periodical includes a volume number, italicize it and then give the page range (in regular type) without "pp." If the periodical does not use volume numbers, as in newspapers, use *p.* or *pp.* for page numbers.

Note: Unlike other periodicals, *p.* or *pp.* precedes page numbers for a newspaper reference in APA style.

Examples:

Harlow, H. F. (1983). Fundamentals for preparing psychology journal articles. *Journal of Comparative and Physiological Psychology*, 55, 893-896.

Henry, W. A., III. (1990, April 9). Making the grade in today's schools. *Time*, 135, 28-31.

Kalette, D. (1986, July 21). California town counts town to big quake. *USA Today*, 9, p. A1.

Kanfer, S. (1986, July 21). Heard any good books lately? *Time*, 113, 71-72.

Trillin, C. (1993, February 15). Culture shopping. *New Yorker*, pp. 48-51.

D. Electronic Sources

Website or Webpage

Format:

Online periodical:

Author's name. (Date of publication). Title of article. *Title of Periodical*, volume number,
Retrieved month day, year, from full URL

Online document:

Author's name. (Date of publication). *Title of work*. Retrieved month day, year, from full URL

Note: When citing Internet sources, refer to the specific website document. If a document is undated, use "n.d." (for no date) immediately after the document title.

Break a lengthy URL that goes to another line after a slash or before a period. Continually check your references to online documents. There is no period following a URL.

Note: If you cannot find some of this information, cite what is available.

Examples:

Devitt, T. (2001, August 2). Lightning injures four at music festival. *The Why? Files*. Retrieved January 23, 2002, from <http://whyfiles.org/137lightning/index.html>

Dove, R. (1998). Lady freedom among us. *The Electronic Text Center*. Retrieved June 19, 1998, from Alderman Library, University of Virginia website: <http://etext.lib.virginia.edu/subjects/afam.html>

Note: If a document is contained within a large and complex website (such as that for a university or a government agency), identify the host organization and the relevant program or department before giving the URL for the document itself. Precede the URL with a colon.

Fredrickson, B. L. (2000, March 7). Cultivating positive emotions to optimize health and well-being. *Prevention & Treatment*, 3, Article 0001a. Retrieved November 20, 2000, from <http://journals.apa.org/prevention/volume3/pre0030001a.html>
GVU's 8th WWW user survey. (n.d.). Retrieved August 8, 2000, from <http://www.cc.gatech.edu/gvu/usersurveys/survey1997-10/>

Health Canada. (2002, February). *The safety of genetically modified food crops*. Retrieved March 22, 2005, from http://www.hc-sc.gc.ca/english/protection/biologics_genetics/gen_mod_foods/genmodebk.html

Hilts, P. J. (1999, February 16). In forecasting their emotions, most people flunk out. *New York Times*. Retrieved November 21, 2000, from <http://www.nytimes.com>

10. Appendices

Appendices shall be provided for supplementary materials that are not appropriate for inclusion in the text.

10.1 Number the appendices using Arabic numerals, e.g., Appendix 1, Appendix 2, etc

10.2 Type the appendix title centered and triple-spaced from the top margin.

10.3 If lengthy, supplementary illustrative materials such as tables, figures, and references shall appear in appendices and be referred to at appropriate in the text.

13.4 List the appended work/document in the Table of Contents as it is numbered and entitled in the appendices, e.g., Appendix 1: Research Instrument.

III. Binding Specifications and Submission of Manuscripts

The color of the cover of all hardbound theses/dissertations shall be **maroon**.

The spine shall contain the BEC initials, the course of the researchers, the research title, and the year of completion. (*See Appendix 4*)

The Dean shall set the deadline for the submission of the final copies for the course programs. Diplomas, transcripts of records, certifications, and any other pertinent academic documents shall be released only to graduates who have complied with these academic requirements.

References

American Psychological Association. (2010). *Publication manual of the American Psychological Association* (6th ed.). Washington DC: American Psychological Association.

Graduate School. (2007). *Research Policy on thesis and dissertation*. Sta. Mesa, Manila: Polytechnic University of the Philippines. www.collab.itc.virginia.edu

COVER PAGE FORMAT

(Space between the first line of the title and the top edge of the cover shall be 6 inches)

RESEARCH TITLE

(Typewritten in inverted pyramid style, Times New Roman and ALL CAPS)

(8 spaces)

**Batangas Eastern Colleges
College Department**

(10 single spaces)

NAME

(2 spaces)

Degree Program

(8 spaces)

Year

*(Noter All entries shall be typewritten in L4-paint Times New Roman font and in a bold face.
Only the research title and the name of the researchers shall be written in ALL CAPS.*

TITLE PAGE FORMAT

(The first line of the research title must be typed 6 spaces from the paper's top margin.)

RESEARCH TITLE *(Typewritten in
on inverted pyramid style 14-point, Times New Roman
and ALL CAPS)*

(5 spaces)

**A Capstone Project
Presented to the Faculty of the Batangas Eastern Colleges
College Department
San Juan, Batangas**

(6 spaces)

**In Partial Fulfilment of the Requirements for the Degree
Bachelor of Science in Information System**

(8 spaces)

by

(4 spaces)

Name of Researcher (bold face)

(2 spaces)

Month and Year

(NOTE: Except for the title, all entries shall be typed using 12-font)

Appendix 3
CERTIFICATION AND APPROVAL SHEET

This research entitled **"TITLE OF THE STUDY, ALL CAPS, BOLD FACE"** prepared and submitted by **NAME OF RESEARCHERS, ALL CAPS, BOLD FACE** has been approved and accepted in partial fulfillment of the requirements for the degree **State the Degree/ Program in Bold Face and in Proper Capitalization.**

NAME OF THE ADVISER, Honorifics
Adviser

ORAL EXAMINATION COMMITTEE

Approved by the Committee on Oral Examination with a grade of _____.

FULL NAME, Honorifics
Member

FULL NAME, Honorifics
Member

FULL NAME, Honorifics
Chairperson

Accepted and approved in partial fulfillment of the requirements for the degree State the Degree/ Program in Proper Capitalization.

FULL NAME, Honorifics
Dean of Colleges
Date: _____

Appendix 4

ABSTRACT

Title : RESEARCH TITLE HERE

Researchers : Researcher 1

Researcher 2

Researcher 3

Researcher 4

Course : Bachelor of Science in _____

Adviser : First Name. M.I. Lastname, Honorifics

Date : June 2024

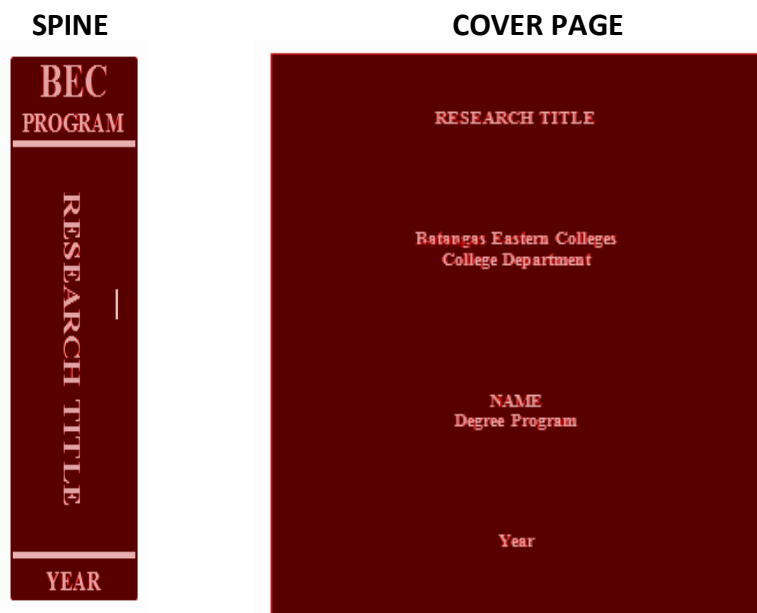
The abstract should not be less than 200 words and not more than 250 words, and should be typed double-spaced and preferably with two to three keywords. Normally the abstract does not include any reference to the literature.

Keywords: Keyword 1, Keyword 2, Keyword 3

Appendix 5 SPINE FORMAT and COLOR

Spine format: The number of bars on the spine represents the level of the program—one for the undergraduate, two for the master's, and three for the doctoral.

Color: Maroon for all programs.



Prepared by:	Checked/Verified by:	Reviewed by:	Approved by:
MICHELLE A. DINO, MIT	Atty. PIO JAY S. CALINGASAN	RENALYN B. ALMARIO, PhD	LORELIE A. GALIT, Ph.D
Chairperson, CCS	Assistant Dean of Colleges	Dean of Colleges	Vice President for Academic Affairs
Date Signed: _____	Date Signed: _____	Date Signed: _____	Date Signed: _____